

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 17

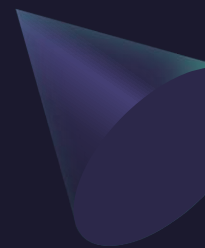


Any Reminders? Any Questions?

- Exam on Friday (discussion) covers 15.1-15.3 + 15.6-15.9
- See email message with resources
 - SP26 Practice Exam/Problems!
- Review during Thursday's class
- Office hours Thursday 3:30-4:30 at MONT 233
- Calc Night Thursday 6:30-8:30 at MONT 104



ALVARO: Start the recording!



“Calculus 3”

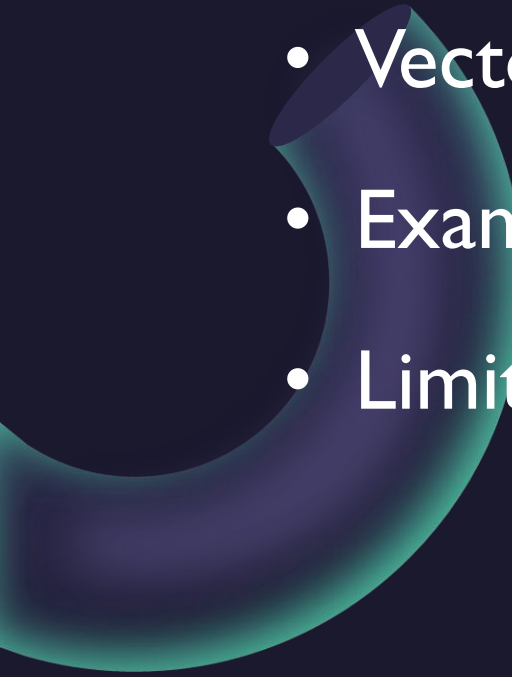
Multi-Variable Calculus

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Vector Functions and Space Curves



Today – Vector Functions and Space Curves

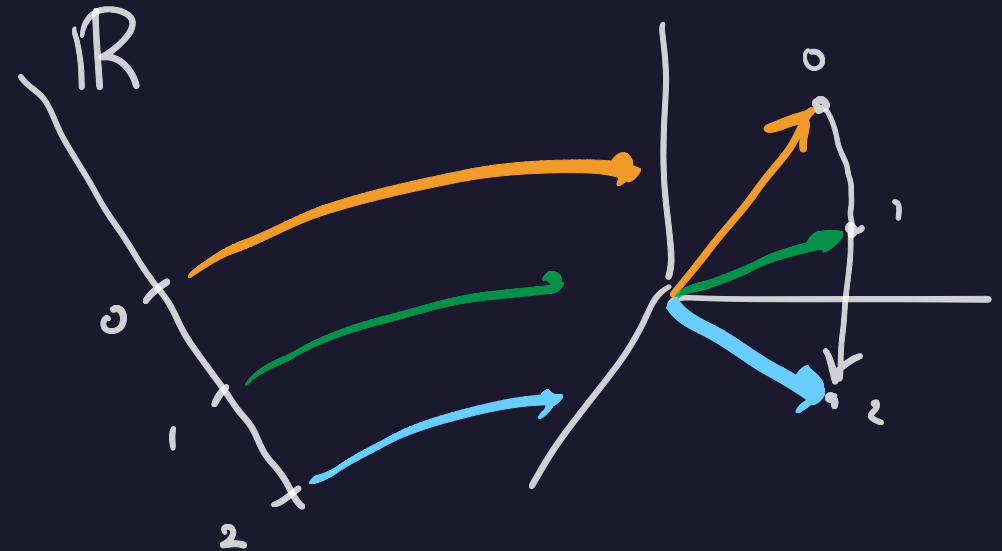
- 
- Vector valued functions
 - Examples
 - Limits and Continuity

What is a “Vector-valued Function”?

$$f: \mathbb{R} \longrightarrow \mathbb{R}^3$$

$$t \longmapsto (f(t), g(t), h(t))$$

$$f: [a, b] \longrightarrow \mathbb{R}^3$$



Example: Describe the image of the function given by

$$r(t) = (1-t, 2, -1+t)$$

$$r: \mathbb{R} \rightarrow \mathbb{R}^3$$

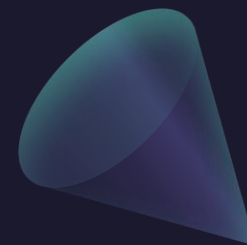
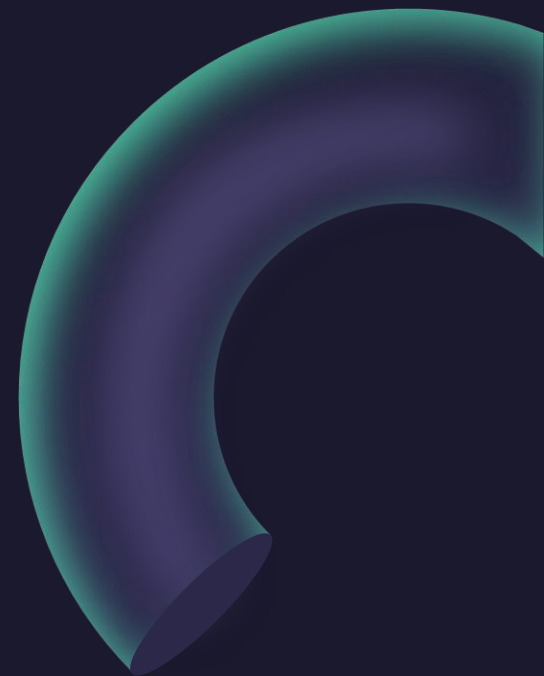
$$t \mapsto (1-t, 2, -1+t)$$

vector form

param
form

$$\begin{cases} x = 1-t \\ y = 2 \\ z = -1+t \end{cases}$$

$$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$



Example: Describe the image of the function given by

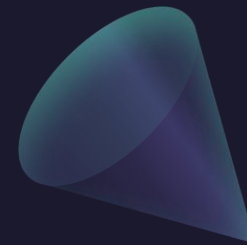
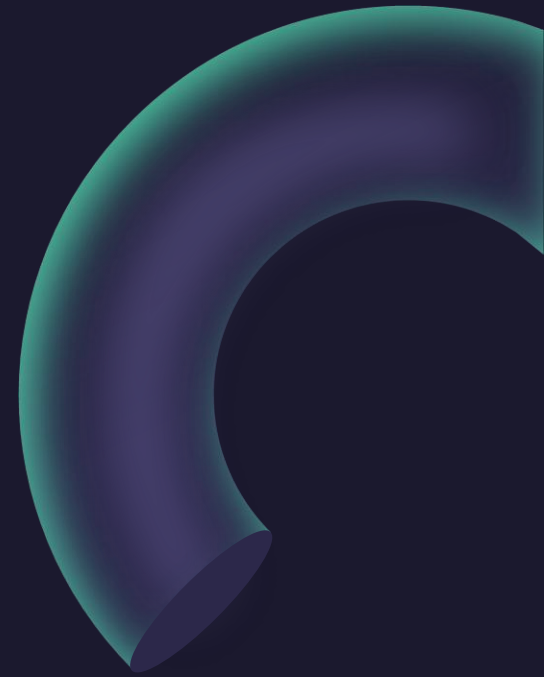
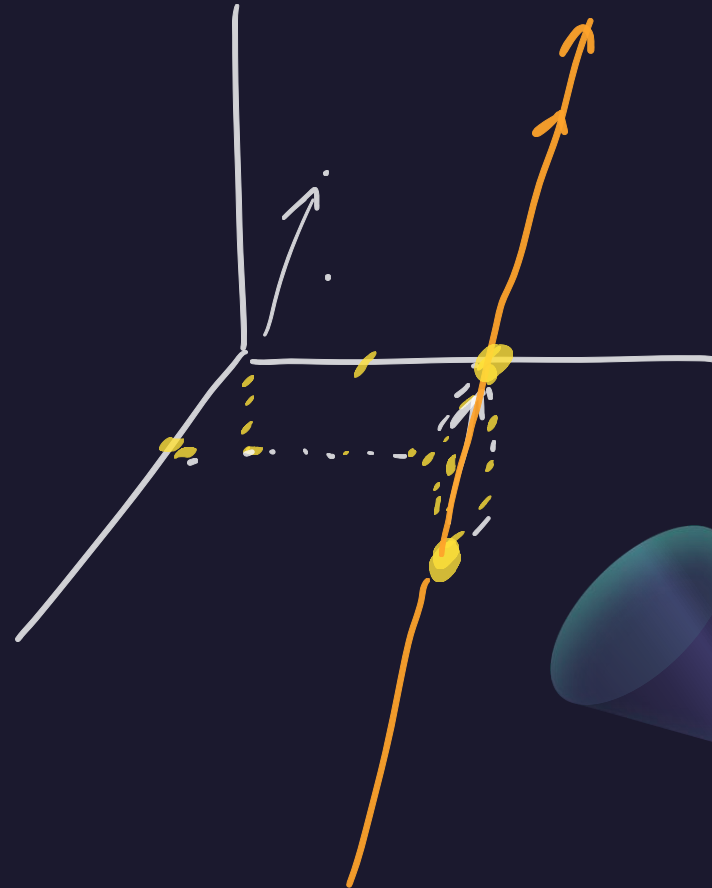
$$r(t) = (1+t, 2+5t, -1+6t)$$

$$(1-t, 2, -1+t)$$

$$\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + t \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$$

$$r(0) = (1, 2, -1)$$

$$r(1) = (0, 2, 0)$$



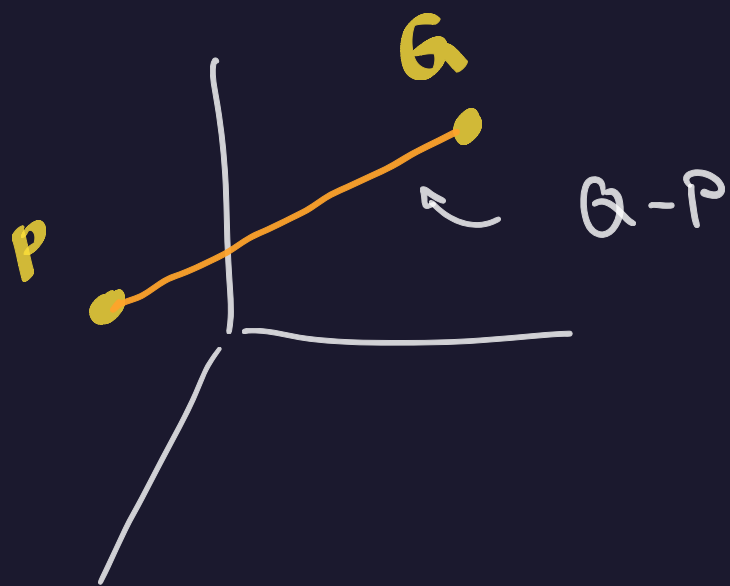
Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

vector
valued
function

$$r(t) = (1-t)P + tQ$$

$$r: [0, 1] \rightarrow \mathbb{R}^3$$

0 $\mapsto$ P
1 $\mapsto$ Q



$$= P + t(Q-P)$$

$$= \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$$

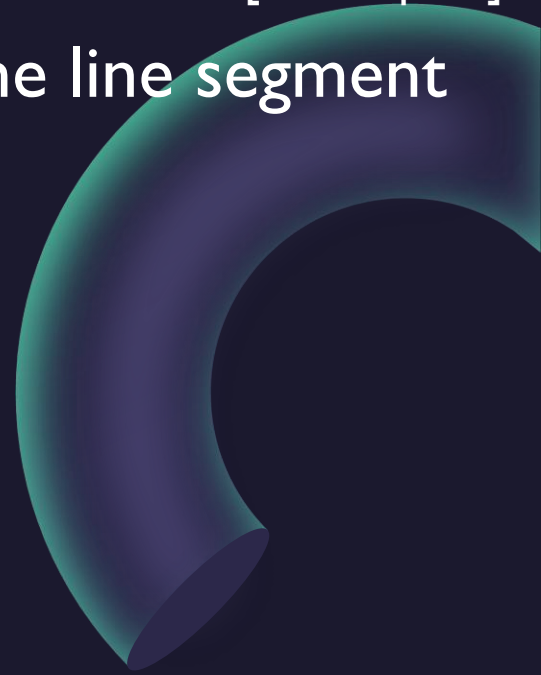
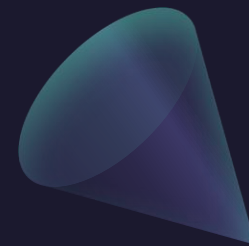
or

$$\begin{cases} x = 1 + t \\ y = -3 + 2t \\ z = 2 + t \end{cases}$$

vector
eq'n form

parametric
form

Example: Find a vector equation and parametric equation for the line segment that joins $P = (1, -3, 2)$ and $Q = (2, -1, 3)$.

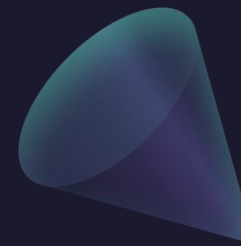
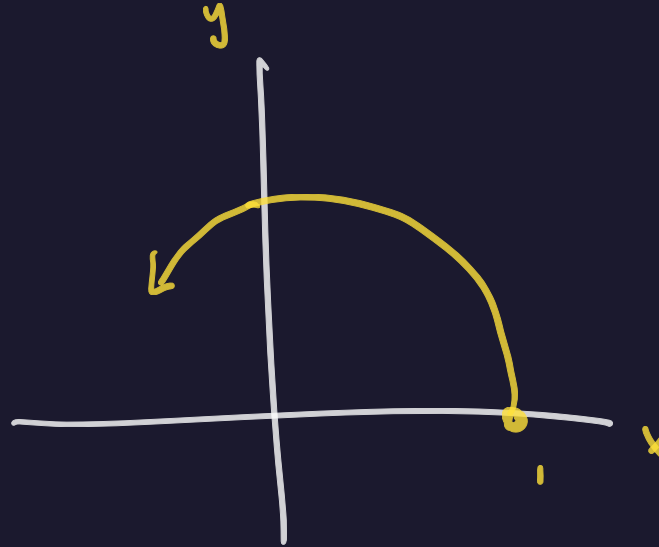


Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

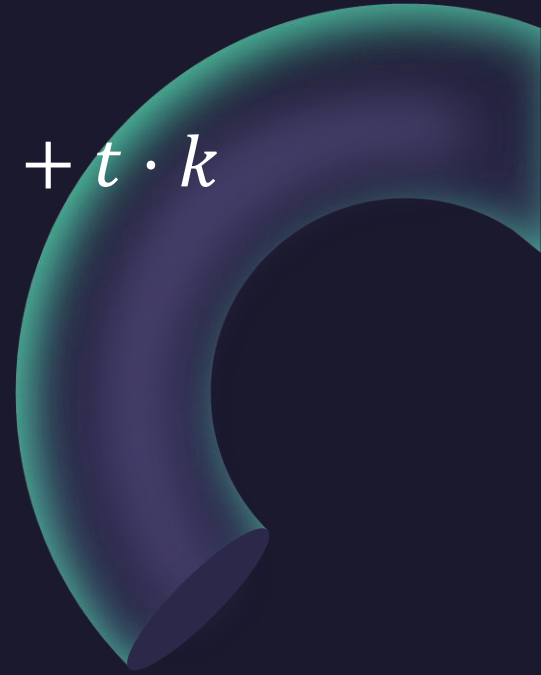
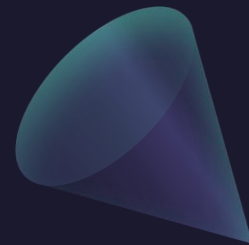
Projection onto xy -plane:

$$\tilde{r}(t) = (\cos t, \sin t)$$

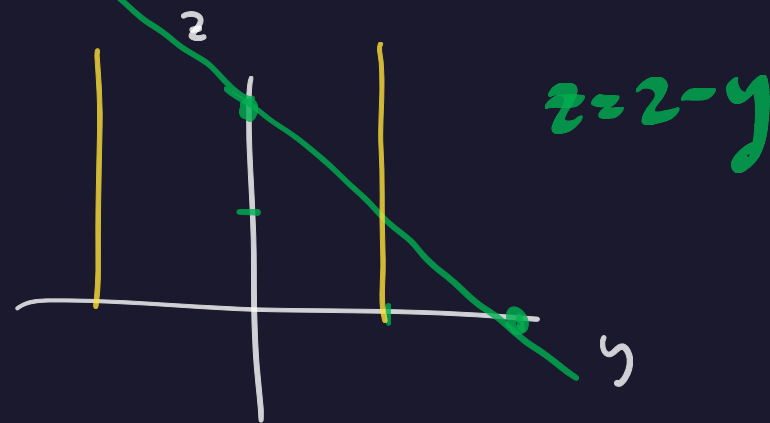


Example: Describe the image of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$



Example: Find a vector equation that represents the curve at the intersection of $x^2 + y^2 = 1$ and the plane $y + z = 2$.

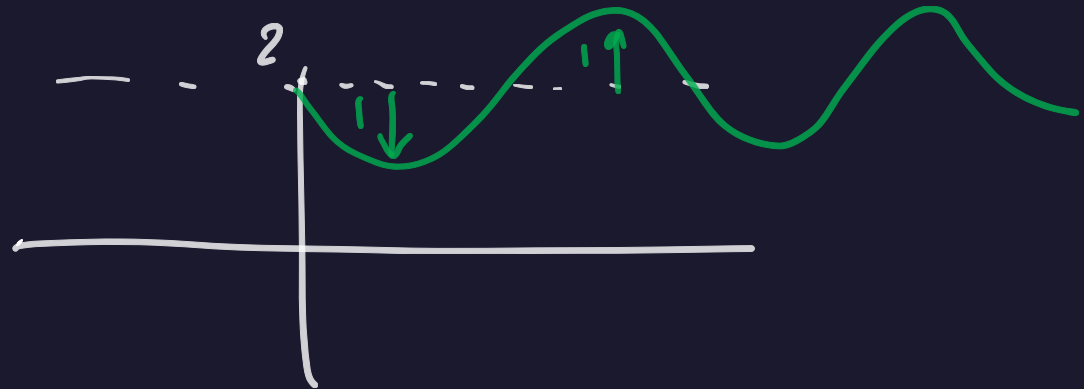


1) parametrize $x^2 + y^2 = 1$
2) $y + z = 2 \rightarrow z = 2 - y$
 $(\cos t, \sin t)$ parametrizes $x^2 + y^2 = 1$
 $\Rightarrow \boxed{(\cos(t), \sin(t), 2 - \sin t)}$

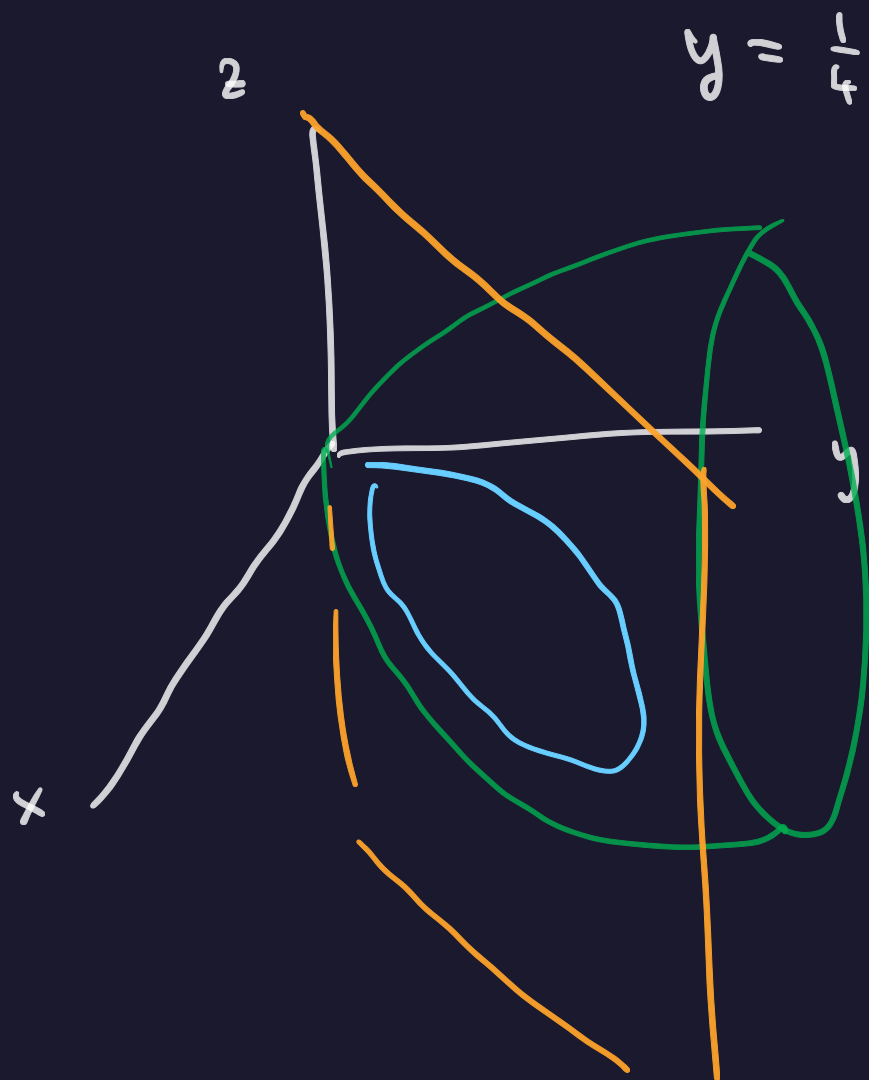
Example: Find a vector equation that represents the curve at the intersection of $x^2 + y^2 = 1$ and the plane $y + z = 2$.

$$\begin{cases} x^2 + y^2 = 1 \\ y + z = 2 \end{cases} \longleftrightarrow \begin{cases} x^2 + y^2 = 1 \\ z = 2 - y \end{cases} \longleftrightarrow (\cos t, \sin t)$$

$$\begin{cases} x = \cos t \\ y = \sin t \\ z = 2 - \sin t \end{cases}$$



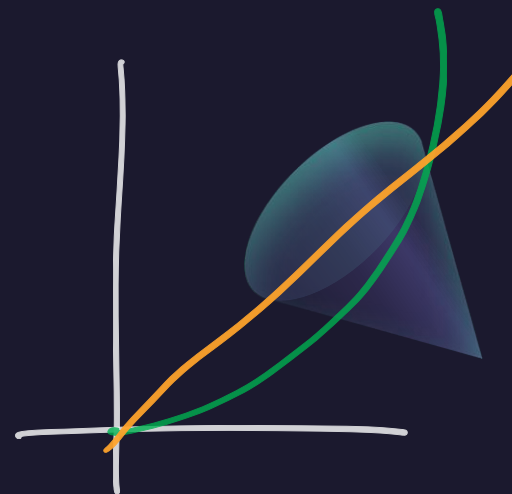
Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.



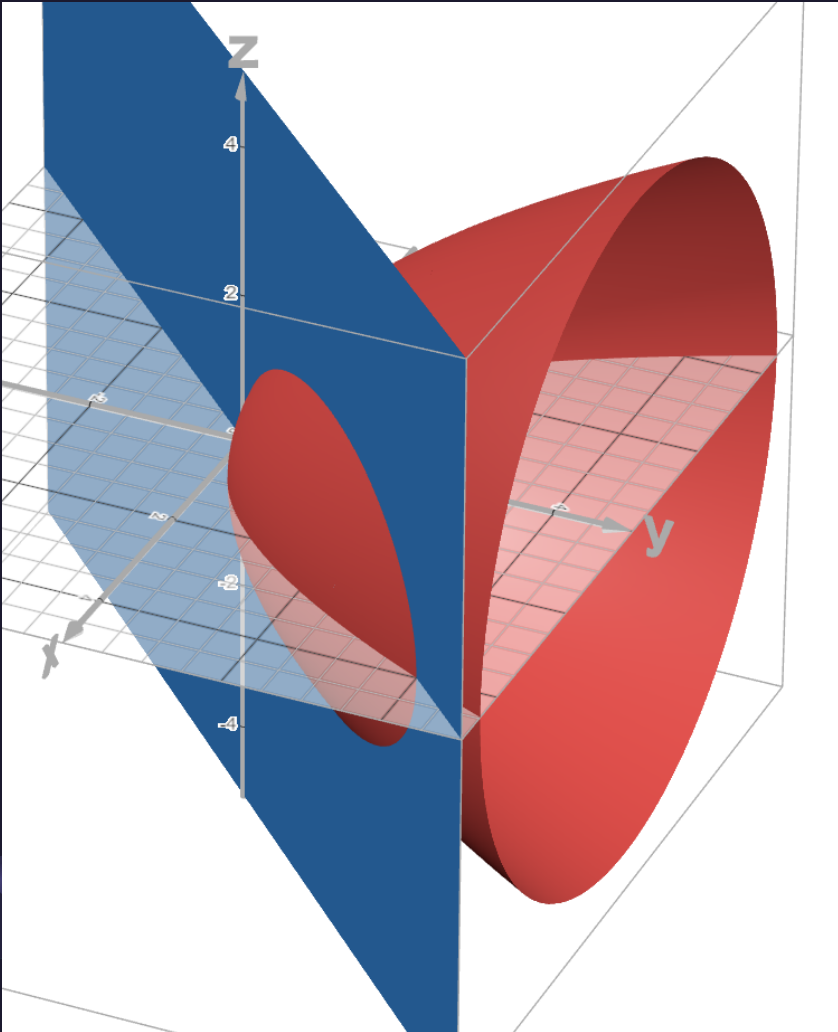
$z=0$

$y = \frac{x^2}{4}$

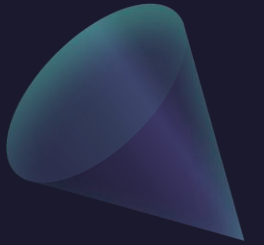
$y = x$



Example: Find a vector equation that represents the curve at the intersection of $4y = x^2 + z^2$ and the plane $y = x$.



$$\begin{cases} 4y = x^2 + z^2 \\ y = x \end{cases} \quad \begin{cases} 4x = x^2 + z^2 \\ y = x \end{cases}$$



$$\begin{aligned} 4x &= x^2 + z^2 \rightarrow x^2 - 4x + z^2 = 0 \\ x^2 - 4x + 4 + z^2 &= 4 \\ (x-2)^2 + z^2 &= 4 \end{aligned}$$

$$\begin{aligned} x-2 &= 2\cos t \\ z &= 2\sin t \end{aligned}$$

$$y = 2 + 2\cos t$$

$$\boxed{r(t) = (2 + 2\cos t, 2 + 2\cos t, 2\sin t)}$$

Limits and Continuity of Vector-valued Functions

1 If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$, then

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \left\langle \lim_{t \rightarrow a} f(t), \lim_{t \rightarrow a} g(t), \lim_{t \rightarrow a} h(t) \right\rangle$$

provided the limits of the component functions exist.

A vector function $\mathbf{r}$ is **continuous at a** if

$$\lim_{t \rightarrow a} \mathbf{r}(t) = \mathbf{r}(a)$$

Example: Find the limit as t goes to 0:

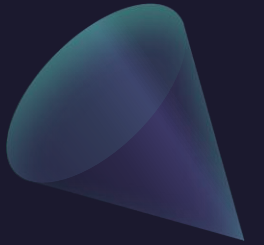
$$\lim_{t \rightarrow 0} \left(1 + t^3, te^{-t}, \frac{\sin(t)}{t} \right)$$

$$= \left(\lim_{t \rightarrow 0} 1 + t^3, \lim_{t \rightarrow 0} te^{-t}, \lim_{t \rightarrow 0} \frac{\sin t}{t} \right)$$

$$= (1, 0, 1)$$

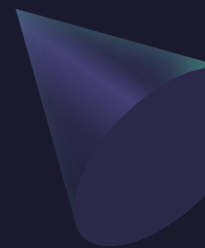
$$\left\{ te^{-t} = \frac{t}{e^t} \rightarrow 0 \right\}$$

Questions?

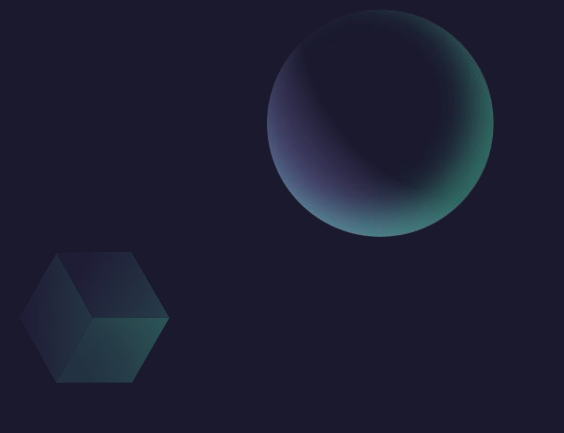




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“Calculus 3”

A sphere, a cube, and a cone are positioned in the upper right area of the slide. They are rendered with a blue-to-teal gradient and soft shadows, giving them a three-dimensional appearance.

Multi-Variable Calculus

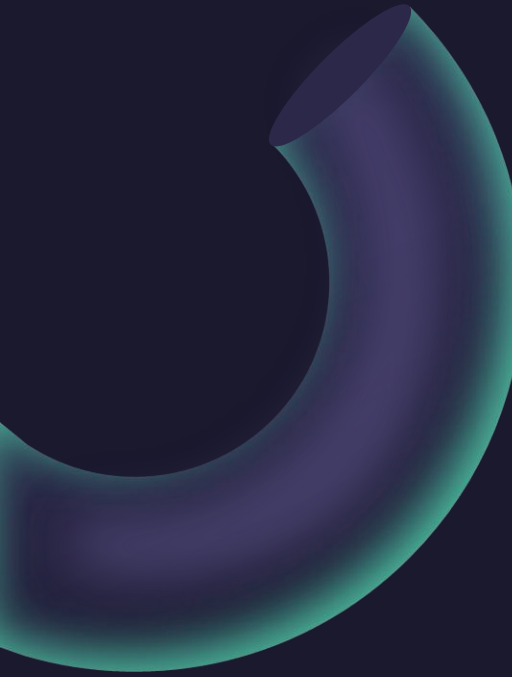
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Derivatives and Integrals Of Vector Functions

A cone and a large torus are positioned in the lower right area of the slide. They are rendered with a blue-to-teal gradient and soft shadows, giving them a three-dimensional appearance.

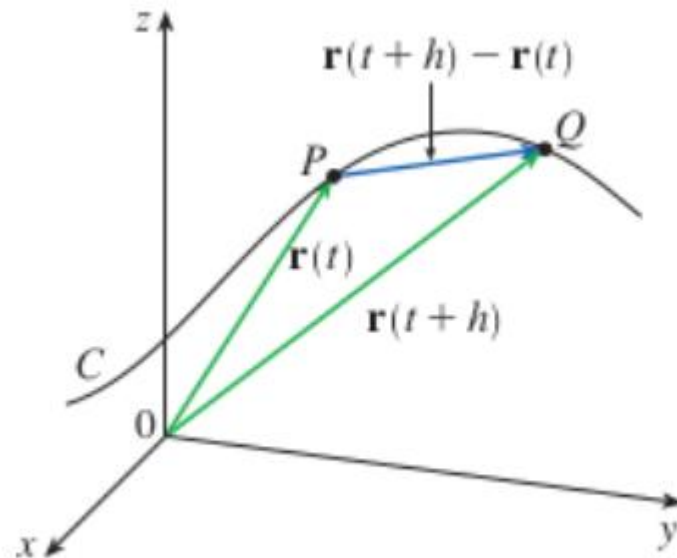
A decorative graphic consisting of a translucent sphere and a cube, both with a teal-to-purple gradient, positioned to the left of the title.

Today – Derivatives and Integrals of Vector Functions

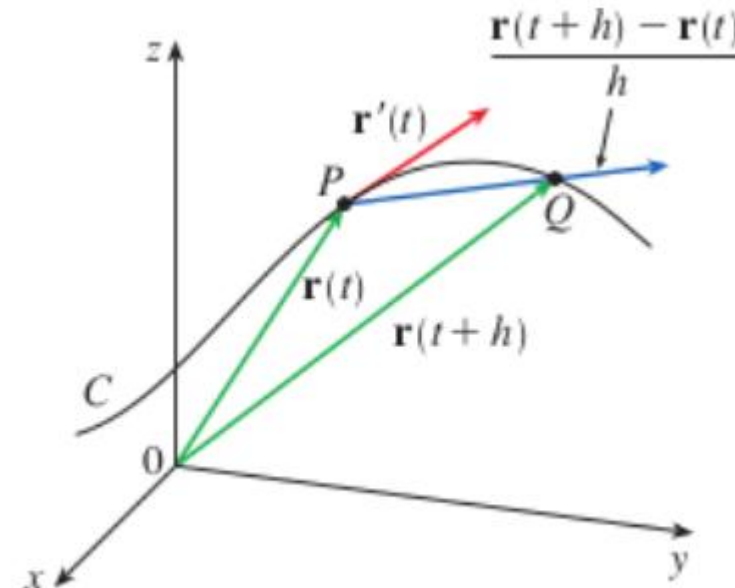
- 
- A decorative graphic of a torus (donut shape) with a teal-to-purple gradient, positioned to the left of the list.
- Derivatives of vector valued functions
 - Differentiation rules
 - Integrals

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$



(a) The secant vector $\overrightarrow{PQ}$



(b) The tangent vector $\mathbf{r}'(t)$

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

If $\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$, where f , g , and h are differentiable functions, then

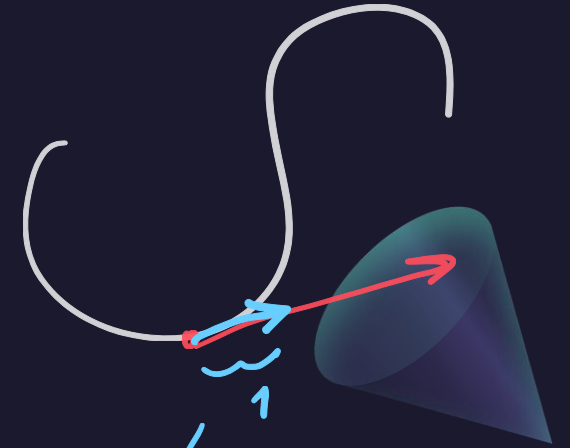
$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle = f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}$$

What is a derivative of a “Vector-valued Function”?

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h \rightarrow 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h}$$

A unit vector that has the same direction as the tangent vector is called the **unit tangent vector $\mathbf{T}$** and is defined by

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$



Example: Find the derivative of the function given by

$$r(t) = (1 + t, 2 + 5t, -1 + 6t)$$

and find the unit tangent vector at $t = 2$.

$$\frac{d}{dt} r(t) = r'(t) = (1, 5, 6)$$

$$\sqrt{1^2 + 5^2 + 6^2} = \sqrt{62}$$

$$r'(2) = (1, 5, 6)$$

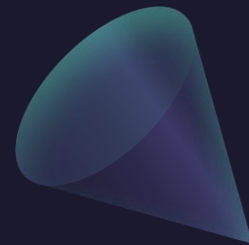
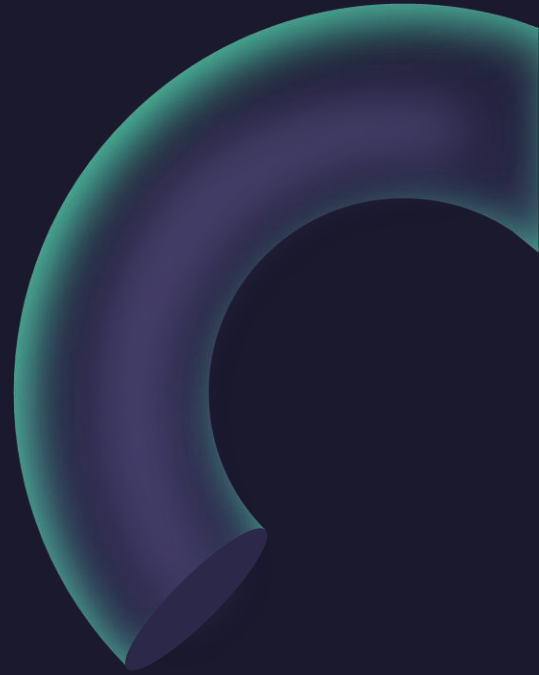
$$T(2) = \frac{r'(2)}{|r'(2)|} = \frac{1}{\sqrt{62}} (1, 5, 6)$$

$$= \left(\frac{1}{\sqrt{62}}, \frac{5}{\sqrt{62}}, \frac{6}{\sqrt{62}} \right)$$

Example: Find the derivative of the function given by

$$r(t) = (1 + t, 2 + 5t, -1 + 6t)$$

and find the unit tangent vector at $t = 2$.



Example: Find the derivative of the function given by

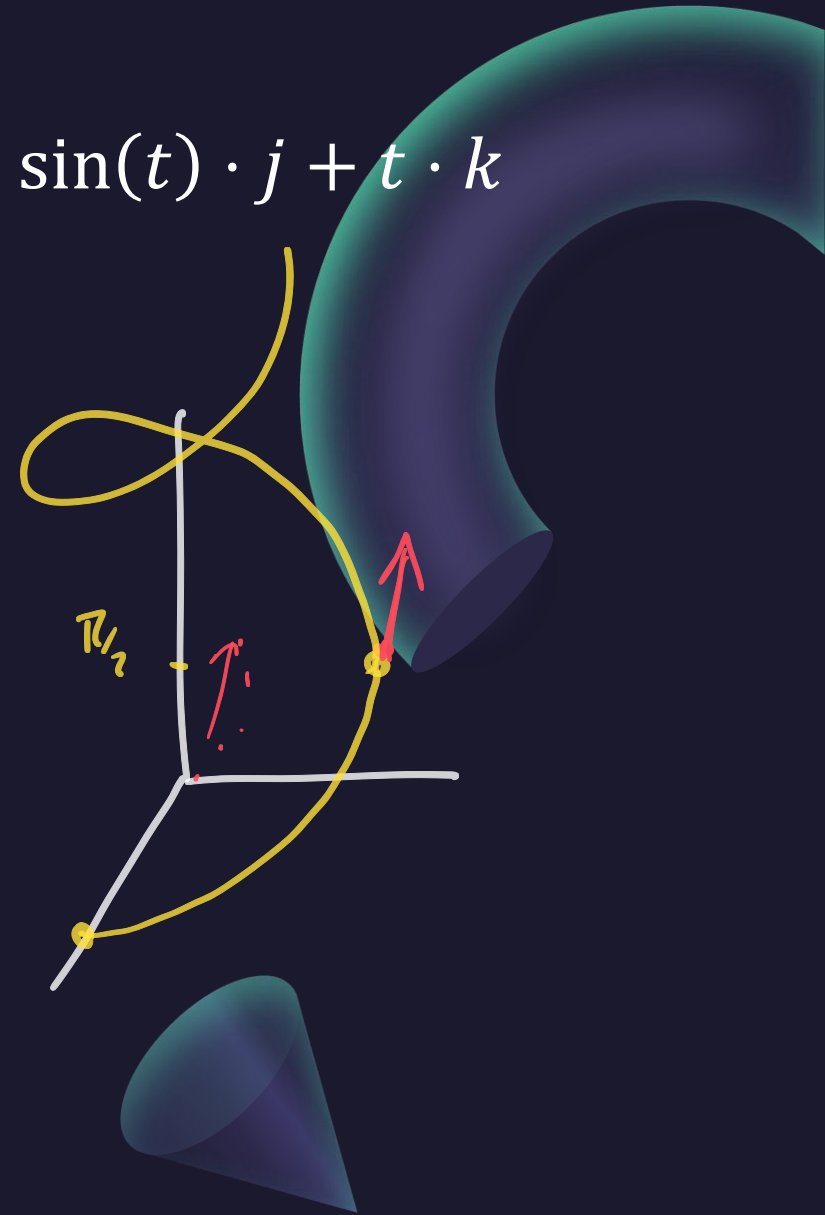
$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$
and find the unit tangent vector at $t = \frac{\pi}{2}$.

$$r'(t) = (-\sin(t), \cos(t), 1)$$

$$t = \pi/2 \quad r'(\pi/2) = (-1, 0, 1)$$

$$r(\pi/2) = (0, 1, \pi/2)$$

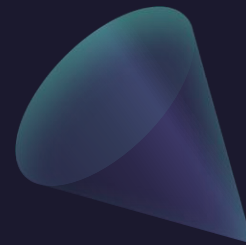
$$T(\pi/2) = \frac{r'(\pi/2)}{|r'(\pi/2)|} = \left(-\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}\right)$$



Example: Find the derivative of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

and find the unit tangent vector at $t = \frac{\pi}{2}$.



Differentiation Rules for Vector-valued Functions

Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function.

Then

$$1. \frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$$

$$2. \frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$$

$$3. \frac{d}{dt}[f(t) \mathbf{u}(t)] = f'(t) \mathbf{u}(t) + f(t) \mathbf{u}'(t) \leftarrow \text{Prod Rule}$$

$$4. \frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t) \quad \text{Dot Product!}$$

$$5. \frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$$

$$6. \frac{d}{dt}[\mathbf{u}(f(t))] = f'(t) \mathbf{u}'(f(t)) \quad (\text{Chain Rule})$$

If $|\mathbf{r}(t)| = c$ (a constant), then $\mathbf{r}'(t)$ is orthogonal to $\mathbf{r}(t)$ for all t .

Integrals of “Vector-valued Functions”

$$\begin{aligned}\int_a^b \mathbf{r}(t) dt &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mathbf{r}(t_i^*) \Delta t \\ &= \lim_{n \rightarrow \infty} \left[\left(\sum_{i=1}^n f(t_i^*) \Delta t \right) \mathbf{i} + \left(\sum_{i=1}^n g(t_i^*) \Delta t \right) \mathbf{j} + \left(\sum_{i=1}^n h(t_i^*) \Delta t \right) \mathbf{k} \right]\end{aligned}$$

and so

$$\int_a^b \mathbf{r}(t) dt = \left(\int_a^b f(t) dt \right) \mathbf{i} + \left(\int_a^b g(t) dt \right) \mathbf{j} + \left(\int_a^b h(t) dt \right) \mathbf{k}$$

Example: Find the integral

$$\int_0^{\pi/2} r(t) dt$$

of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

$$\int_0^{\pi/2} r(t) dt = \left(\int_0^{\pi/2} \cos t dt, \int_0^{\pi/2} \sin t dt, \int_0^{\pi/2} t dt \right)$$

$$= \left(\sin t \Big|_0^{\pi/2}, -\cos t \Big|_0^{\pi/2}, \frac{t^2}{2} \Big|_0^{\pi/2} \right)$$

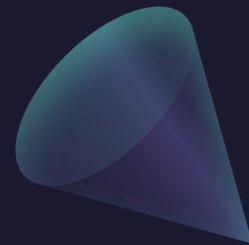
$$= \left(1, 1, \frac{(\pi/2)^2}{2} \right)$$

Example: Find the integral

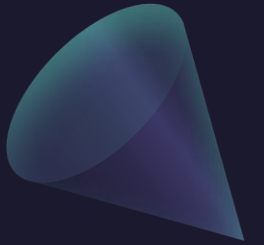
$$\int_0^{\pi/2} r(t) \, dt$$

of the function given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

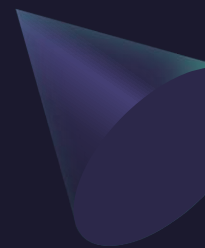


Questions?





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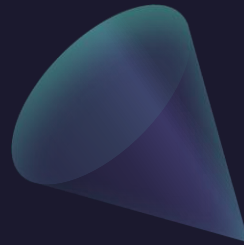
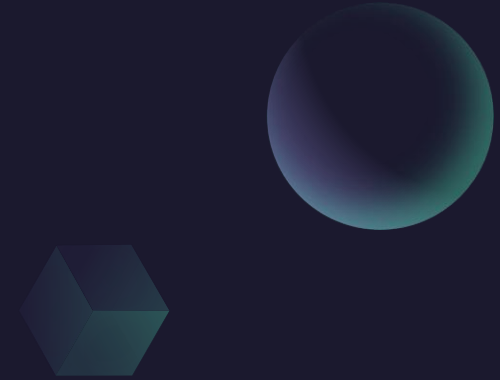


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Multi-Variable Calculus

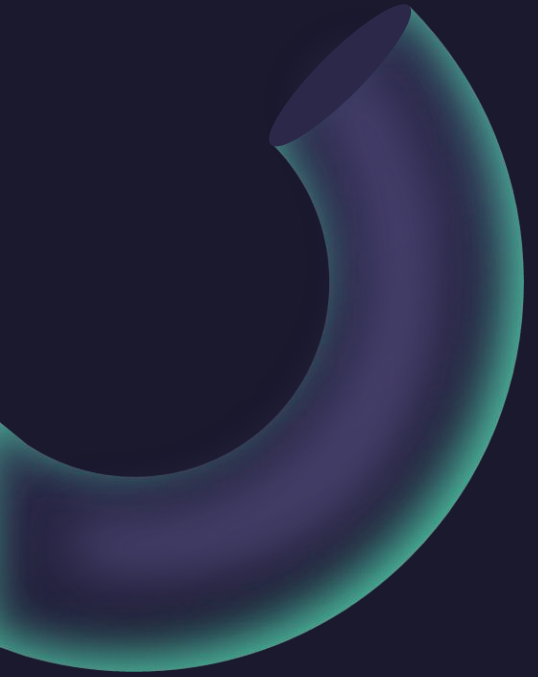
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Arc Length

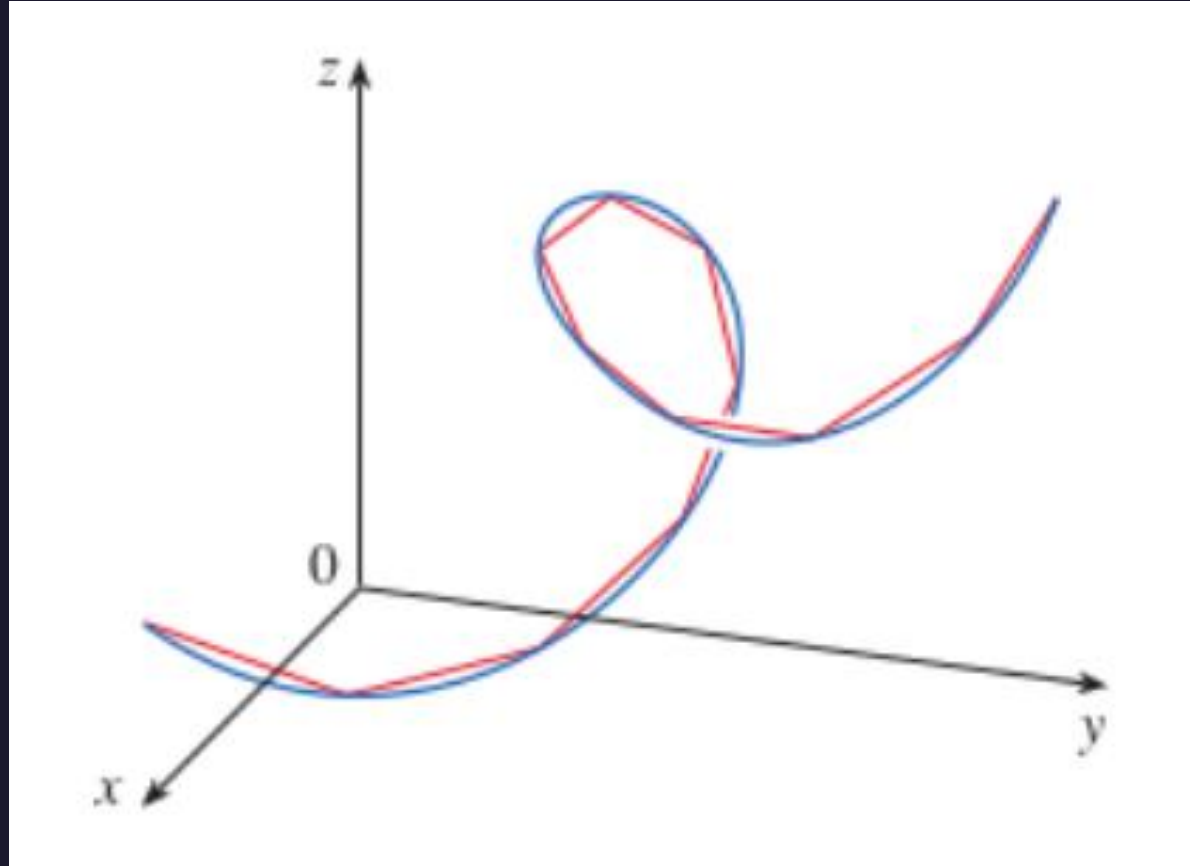


A decorative graphic consisting of a sphere and a cube, both rendered with a blue-to-teal gradient and a subtle glow effect, positioned in the upper-left corner of the slide.

Today – Arc Length

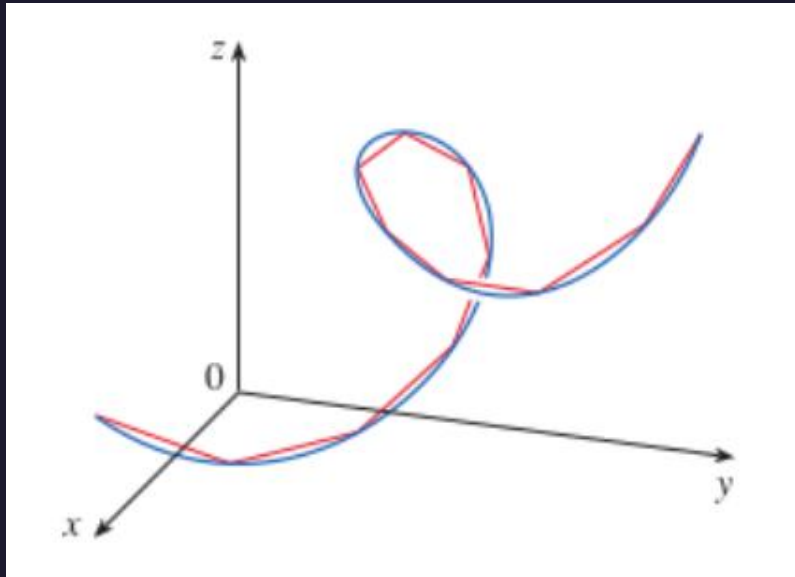
- 
- A decorative graphic of a torus (donut shape) rendered with a blue-to-teal gradient and a subtle glow effect, positioned in the lower-left corner of the slide.
- Length of a segment of a curve...
 - In the plane
 - In 3d-space

How do we compute the length of a curve?



How do we compute the length of a curve?

$$\mathbf{r}(t) = (x(t), y(t), z(t))$$



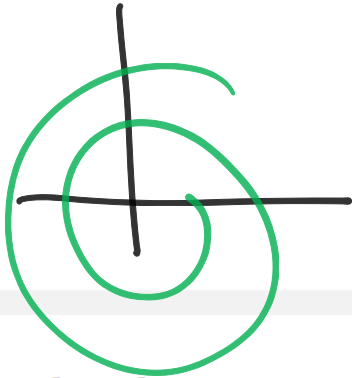
$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

$$\begin{aligned} L &= \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2} dt \\ &= \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt \end{aligned}$$

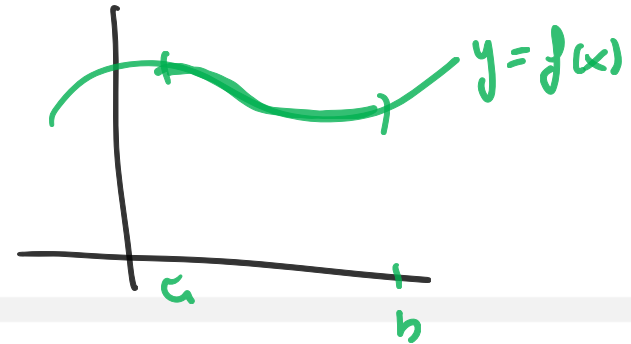
$$\mathbf{r}'(t) = (x', y', z')$$

$$|\mathbf{r}'(t)| = \sqrt{x'^2 + y'^2 + z'^2}$$

How do we compute the length of a curve?



$$L = \int_a^b |\mathbf{r}'(t)| dt$$



because, for plane curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j}$,

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j}| = \sqrt{[f'(t)]^2 + [g'(t)]^2}$$

$$\mathbf{r}(t) = (t, f(t))$$

$$\mathbf{r}' = \left(1, \frac{df}{dt}\right)$$

and for space curves $\mathbf{r}(t) = f(t) \mathbf{i} + g(t) \mathbf{j} + h(t) \mathbf{k}$,

$$\int_a^b \sqrt{1^2 + \left(\frac{df}{dt}\right)^2} dt$$

$$|\mathbf{r}'(t)| = |f'(t) \mathbf{i} + g'(t) \mathbf{j} + h'(t) \mathbf{k}| = \sqrt{[f'(t)]^2 + [g'(t)]^2 + [h'(t)]^2}$$

Example: Find the length of the curve given by

$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$
from the point $P = (1, 0, 0)$ to the point $Q = (1, 0, 2\pi)$.

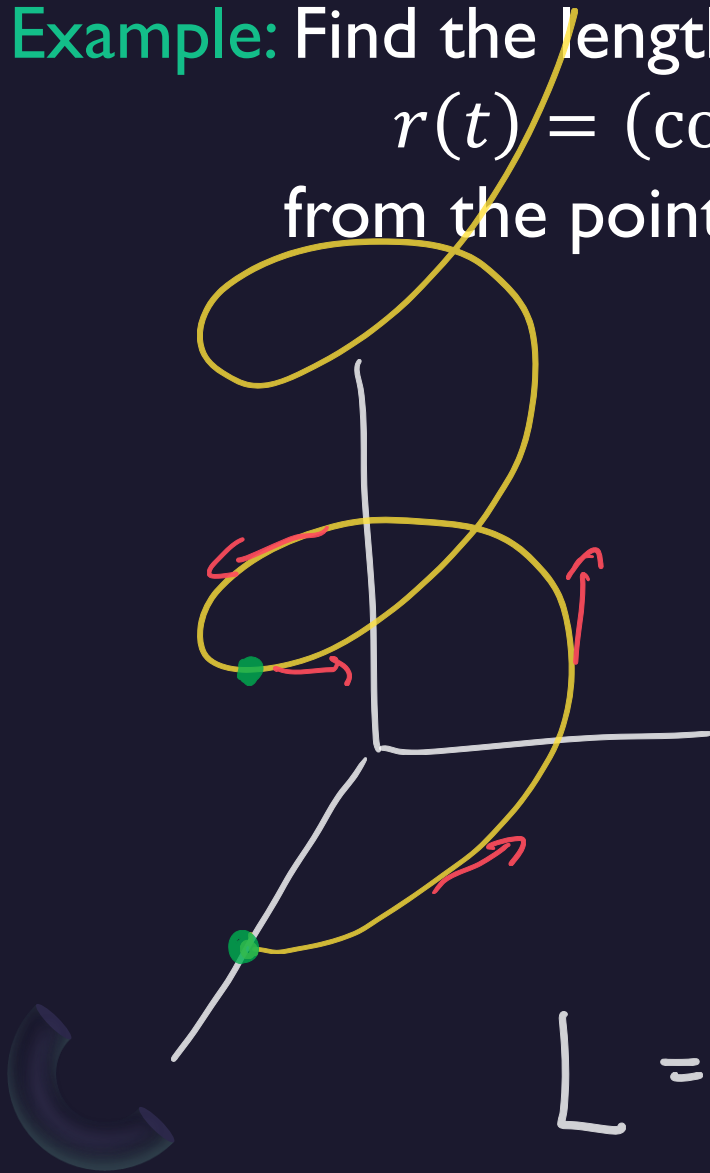
$$t=0 \quad (1, 0, 0)$$

$$t=2\pi \quad (1, 0, 2\pi)$$

$$r'(t) = (-\sin t, \cos t, 1)$$

$$|r'(t)| = \sqrt{(-\sin t)^2 + (\cos t)^2 + 1^2} = \sqrt{2}$$

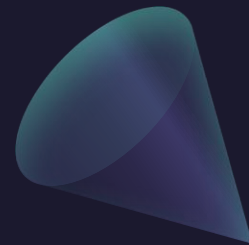
$$L = \int_0^{2\pi} |r'(t)| dt = \int_0^{2\pi} \sqrt{2} dt = \boxed{2\sqrt{2}\pi}$$



Example: Find the length of the curve given by

$$r(t) = (\cos(t), \sin(t), t) = \cos(t) \cdot i + \sin(t) \cdot j + t \cdot k$$

from the point $P = (1, 0, 0)$ to the point $Q = (1, 0, 2\pi)$.



Example: Find the length of the curve given by

$$r(t) = (2t, t^2, \frac{t^3}{3})$$

from $t = 0$ to $t = 1$.

$$r'(t) = (2, 2t, t^2)$$

$$|r'(t)| = \sqrt{2^2 + (2t)^2 + (t^2)^2}$$

$$= \sqrt{4 + 4t^2 + t^4}$$

$$= \sqrt{(t^2 + 2)^2} = t^2 + 2$$

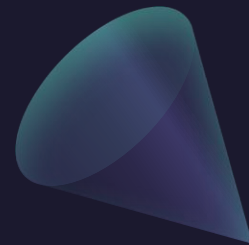
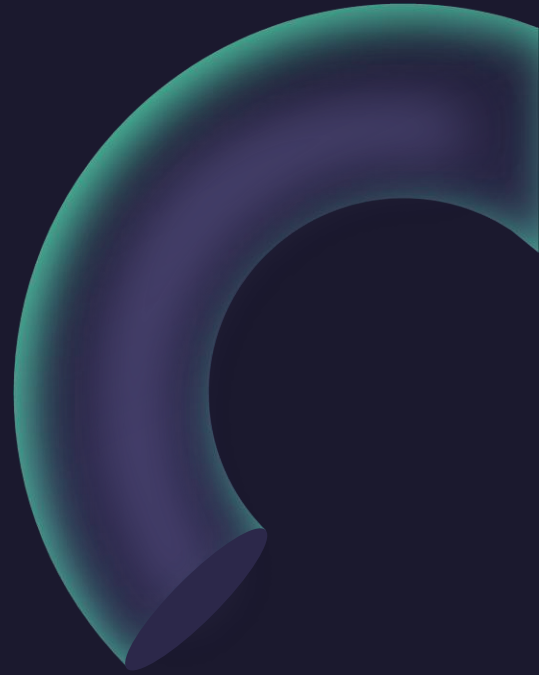
$$L = \int_0^1 |r'(t)| dt$$

$$= \int_0^1 (t^2 + 2) dt = \left. \frac{t^3}{3} + 2t \right|_0^1 = \frac{1}{3} + 2 = \boxed{\frac{7}{3}}$$

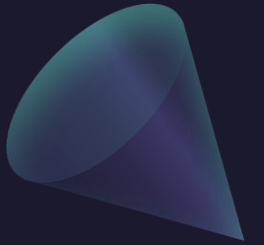
Example: Find the length of the curve given by

$$r(t) = \left(2t, t^2, \frac{t^3}{3}\right)$$

from $t = 0$ to $t = 1$.



Questions?



Thank you

Until next time.

